

International Marketing Analytics-

Course Content and Style



WIRTSCHAFTS
UNIVERSITÄT
WIEN VIENNA
UNIVERSITY OF
ECONOMICS
AND BUSINESS

Dr Arne Floh

Senior Lecturer in International Business

arne.floh@wu.ac.at

Winter term 2026/27



The Journey begins...



Instructor - Dr Arne Floh

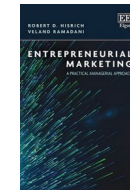
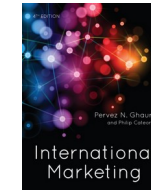
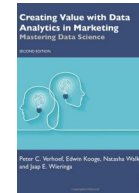


- Personal Information
 - **Senior Lecturer in International Marketing**
 - Institute for International Business, WU Vienna
 - Deputy Program Director Export- und Internationalisierungsmanagement
 - WWW: [Wirtschaftsuniversität Wien: Arne Floh - Faculty and Staff - Team \(wu.ac.at\)](http://www.wirtschaftsuniversitaet-wien.ac.at/faculty-and-staff/team/wu.ac.at)
 - LinkedIn: <http://www.linkedin.com/arnefloh>
- Contact Information
 - T: 01 31336 6367
 - E: arne.floh@wu.ac.at
 - Room D1.5.068
 - Office Hours: by appointment
 - Please contact me first by email!

Instructor - Dr Arne Floh



- Work Experience
 - International & Industry Background
 - Teaching Experience > 20 Years
- Teaching
 - Entrepreneurial Marketing
 - International Marketing
 - Marketing Research and Marketing Analytics
 - Digital Marketing & Social Media
 - Supervision (BSc, MSc, PhD)
- Research
 - Consumer Behaviour, Relationship Marketing
 - Electronic Marketing & Social Media
- Private
 - Football
 - Electronic Gadgets
 - Austrian Wine



- **Marketing analytics turns marketing data into decisions: which markets to invest in, how to price, how much to spend on which channel, and whether it worked.**
- **This module concentrates on the question that dominates international marketing budgets today: how should a fixed media budget be split across countries and channels?**
- **Marketing mix modelling (MMM) has returned to the centre of this debate since privacy changes made user-level tracking unreliable. Managers who can read, challenge and commission these models hold the budget conversation.**
- **We work on one running case throughout: Alpenglöck, a Vienna-based premium chocolate brand active in six European markets (AT, DE, FR, IT, NL, PL).**
- **All analyses are done in Python with AI coding assistants doing most of the typing. You learn to specify, run, check and communicate analyses, not to memorise syntax.**

Module Content (continued)

Each session starts from a decision an international marketing manager has to take and works backwards to the regression that answers it.

Typical questions which will be answered in this module include:

- Is our price too high in Poland? Price and promotion elasticities across countries
- What did TV do for us in Germany? Carryover, saturation and return on ad spend
- Where should the next euro go? Budget allocation across six countries and five channels
- Can we trust the model? Sales forecasts and geo-lift experiments
- Which channel deserves the credit for a sale in each market? Multi-market attribution

Learning Outcomes

On successful completion of this module, students will be able to:

- 1. Explain Marketing Mix Modelling:** Understand the logic of MMM, including carryover (adstock), saturation, baseline versus incremental sales, and ROAS versus marginal ROAS.
- 2. Estimate Market Response Models:** Specify, estimate and diagnose regression models for marketing data in Python, from OLS and log-log elasticity models to panel fixed effects and hierarchical Bayesian MMM.
- 3. Allocate Budgets Across Markets:** Translate model output into a cross-country, cross-channel budget allocation and defend it under uncertainty.
- 4. Forecast Sales:** Produce and evaluate short-term sales forecasts as the baseline for planning.
- 5. Design Experiments:** Design and analyse a geo-lift experiment and use it to validate and calibrate an MMM.
- 6. Compare Attribution Approaches:** Explain why last-touch, regression-based and Shapley attribution disagree, and why results differ across markets.

Learning Outcomes (continued)

- 7. Work Reproducibly:** Use Python, Positron, Quarto and GitHub to build analyses that colleagues can rerun and audit.
- 8. Use AI Assistants Responsibly:** Direct an AI coding assistant, verify its output and document the prompts behind an analysis.
- 9. Enhance Communication Skills:** Communicate analytical findings to a board audience through a written report and a pitch.
- 10. Cultivate Teamwork:** Gain experience in collaborative, version-controlled analytical work and collective data-driven decisions.
- 11. Foster Critical Thinking:** Critically evaluate data sources, model assumptions and findings in the context of international marketing challenges.

Methods of Teaching & Learning

- The course is taught in person as a block of five Tuesday sessions between 6 October and 3 November 2026.
- Every session follows the same rhythm: a concept lecture (60 min), a guided lab in Positron (90 min), a team exercise on the project data (60 min) and a debrief.
- Coding is AI-assisted. You bring your own laptop with Python, Positron, Quarto and GitHub Desktop installed (setup guide on Canvas and GitHub) and an AI assistant of your choice.
- All materials, data and code live in the course GitHub repository. Teams work in their own repositories; pull, commit and push are part of the routine from Session 1.
- It is highly recommended to read the relevant book chapters in advance. Q&A forums on Canvas facilitate the discussion between sessions.

Module Roadmap

Session	Date	Topic	Methods	Chapter
1	Tue 6 Oct	Course Content & Style, Toolkit (Python, Positron, Quarto, GitHub, AI assistants), Regression Foundations	Descriptive Statistics, OLS, Log-log Elasticities, Dummies and Interactions	1, 11
2	Tue 13 Oct	Marketing Mix Modelling I: One Country	Adstock, Saturation, Decomposition, ROAS, Diagnostics	3
3	Tue 20 Oct	Marketing Mix Modelling II: Many Countries, One Budget	Panel Fixed Effects, Hierarchical Bayesian MMM, Budget Optimisation	3
4	Tue 27 Oct	Forecasting, Experiments and Attribution	Time Series Regression, Difference-in-Differences, Logistic Regression, Shapley Values	9, 4
5	Tue 3 Nov	Group Project Pitches and Wrap-up	Board Presentation, Ground-truth Reveal	11

Yildirim, G. and Kübler, R. V., 2025. Applied Marketing Analytics Using Python. SAGE. Times and rooms: see Canvas.

■ **Assessment Strategy**

- Online Certificates Python and GitHub (Individual) – 5 %
- Self-Reflection (Individual) – 5 %
- Online Quizzes (Individual) – $4 \times 5 \% = 20 \%$
- Lab Check-ins (Individual) – $2 \times 10 \% = 20 \%$
- Group Project: Written Report, GitHub Repository and Pitch (Group) – 40 %
- Peer Review – 10 %

■ **Group Enrolment**

- Canvas, groups of 5, by the end of Session 1
- Grades follow the WU scale (1: 90 % and above, 2: 80–89 %, 3: 70–79 %, 4: 60–69 %). Attendance in at least 80 % of sessions is required.

- Take the following free online courses before the second session:
 - Kaggle Learn: Python (about 5 hours, certificate on completion)
 - GitHub Skills: Introduction to GitHub (about 1 hour)
- Install the course environment following the setup guide (Python via uv, Positron, Quarto, GitHub Desktop) and complete the installation check.
- Upload both certificates and a screenshot of the passing installation check on Canvas before the second session (Mon 12 Oct, 11:59 pm).

Online Quizzes (individual)

- Read the session materials (available on Canvas and GitHub)
- Ask questions during the lecture

- Take an online quiz after Sessions 1 to 4
 - Opens on Wednesdays, 9 am
 - Closes on Mondays, 11:59 pm

- Multiple-Choice Questions, closed book, concepts not code

- 10 questions in 10 minutes

Lab Check-ins (individual)

- Each campus session covers a separate topic of international marketing analytics. Please check the previous slide (module roadmap).
- After a short lecture on the substantive topic, a guided lab is held in Positron. Students follow the instructor and use an AI assistant to write and explain the code.
- After Sessions 2 and 4, students replicate the lab for another country or dataset and submit a short Quarto notebook plus the result files.
- Submissions are checked automatically for plausibility (ranges, consistency with the data) and reviewed for interpretation. Each check-in accounts for 10 points.
- Students submit on Canvas: the Quarto file, the rendered HTML, the result files and a short prompt log.

Peer Rating Form

Each group member is required to submit a peer rating form. This form must contain an evaluation (i.e. a grade) of each group member. Do not assign yourself a grade as this would defeat the idea of peer rating. **The form will be kept in strict confidence.**

In the spaces provided below, please fill in your name, choose your group, and write complete names of all of your group members. Finally, assign a grade on a scale from 0 to 10 to each of your group members (**10 being the highest** and 0 being the lowest grade). Please assign grades based on the following criteria:

- → Willingness to carry out assigned tasks
- → Ability to meet deadlines
- → Quality of the individual's work
- → Communication ability and respectful treatment of others
- → Availability for and attendance at group meetings
- → Following the pre-agreed rules

Your Name: → Please enter your name

Your Group: → Choose your group

Group Member Names	Grade
1. Group Member Name	Insert grade
2. Group Member Name	Insert grade
3. Group Member Name	Insert grade
4. Group Member Name	Insert grade
5. Group Member Name	Insert grade

Group Project

- Groups of 5 act as Alpenglow's analytics team and recommend the 2027 media budget split across six countries and five channels.
- Submission files (GitHub repository and Canvas):
 - Quarto report and rendered output (HTML recommended, max 12 pages)
 - allocation_2027.csv: recommended weekly spend per country and channel
 - Pitch deck (max 8 slides) and prompt log
- Milestones: repository and data exploration before Session 2, first model table before Session 4
- Online consulting hours (registration needed)
- **Submission: Mon 2 Nov 2026, 6 pm. Pitches: Session 5, Tue 3 Nov 2026**

Any Questions?